

Undecidability in Uniform Turán Densities

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Abstract

We prove that exact comparison of uniform Turán densities of finite hypergraph families is undecidable in every sufficiently large uniformity. More precisely, there are an effectively computable integer $K_* \geq 4$ and an effectively computable rational number $T \in (0, 1)$ such that, for every $K \geq K_*$, no algorithm can distinguish

$$\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) = T \quad \text{from} \quad \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) > T$$

for finite input families $\mathcal{F}$ promised to have density at least T . Equivalently, equality, non-strict comparison, and strict comparison with this rational threshold are all undecidable under the same promise. For each $K \geq K_*$, we also effectively construct a fixed finite family $\mathcal{G}_K$ of K -graphs with density T ; replacing T by $\pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K)$ gives the same three undecidability statements.

The reduction starts from a fixed finite group presentation extracted from a theorem of Kharlamovich, Myasnikov, and Sapir. We convert its finite-quotient dichotomy into a weighted finite-action estimate, compile finitely many typed palette constraints into ordinary finite forbidden families, and calibrate the resulting density polynomial at a rational point. The palette characterization of Lin, Sun, Wang, and Zhou then transfers the construction to uniform Turán density. Unrestricted middle coordinates extend it to every $K \geq K_*$ without changing the threshold.

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1 Introduction

Uniform Turán density asks how dense a hypergraph can remain under a strong uniformity condition on its edge distribution while avoiding prescribed configurations. For an integer $k \geq 3$, the parameter $\pi_{k-2}^{\text{unif},(k)}(\mathcal{F})$ is the strongest nontrivial member of the corresponding hierarchy for k -graphs. We retain the superscript “unif” to distinguish it from the minimum $(k-2)$ -degree Turán density, which is a different invariant. Erdős and Sós [3] initiated the study of uniform Turán density. Writing $K_4^{(3)-}$ for the unique 3-graph with four vertices and three edges, the equality $\pi_1^{\text{unif},(3)}(K_4^{(3)-}) = 1/4$ was determined by Glebov, Král’, and Volec [6] and by Reiher, Rödl, and Schacht [20]. The zero-density characterization of Reiher, Rödl, and Schacht [19] implies that the uniform Turán density of every single k -graph is either zero or at least k^{-k} . Garbe, Král’, and Lemaire [5] constructed 3-graphs attaining this lower bound, and Lin, Wang, and Zhou [15] did so in every uniformity. See Reiher’s survey [18] for further background.

The randomized construction underlying the palette method goes back to Rödl [21]; Reiher [18] later formulated the palette viewpoint. Lemaire [12] proved that palettes determine the exact uniform Turán density of every single 3-graph, and Král’, Kučerák, Lemaire, and Tardos [11] developed a palette-homomorphism classification. Lin, Sun, Wang, and Zhou [14, Theorem 1.4] proved the form needed here: palettes characterize $\pi_{k-2}^{\text{unif},(k)}(\mathcal{F})$ for every $k \geq 3$ and every family $\mathcal{F}$. They also asked two related membership questions, namely whether a given $\alpha \in [0, 1]$ is the uniform Turán density of a single k -graph or of a finite family. Here the input is instead a finite forbidden family, and the comparison threshold is fixed in advance. We show that this exact comparison problem is undecidable in every sufficiently large uniformity.

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King, Sales, and Schülke [10] related ordinary 3-graph Lagrangians to uniform Turán densities of possibly infinite families and showed that these densities are not well ordered. King, Piga, Sales, and Schülke [9] proved that every palette Lagrangian is attained by a finite forbidden family. Liu and Pikhurko [16] showed that the uniform Turán densities of finite 3-graph families are dense in a nondegenerate interval, while Li, Liu, and Pikhurko [13] obtained a terminal interval for arbitrary 3-graph families. For every $m \geq 2$, Ai, Chen, Chen, Yan, and Zhao determined the sharp directed-triangle m -color palette endpoint $1/3 - 1/(3m^2)$; they realized these values by finite forbidden families and constructed individual 3-graphs whose uniform Turán densities converge to $1/3$ [1]. These results describe the range of possible values. Our results instead concern the algorithmic impossibility of comparing one such value exactly.

Our primary result uses a rational threshold common to all sufficiently large uniformities. Here a decision problem under a promise asks for an algorithm that is correct on every encoded finite family satisfying that promise; its behavior on other inputs is irrelevant.

Theorem 1.1. *There are an effectively computable integer $K_* \geq 4$ and an effectively computable rational number $T \in (0, 1)$ such that, for every integer $K \geq K_*$ and under the promise $\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) \geq T$, each of the relations*

$$\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) = T, \quad \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) \leq T, \quad \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) > T$$

is undecidable for finite input families $\mathcal{F}$.

For every $K \geq K_*$, the reduction constructs, effectively from an input $e \in \mathbb{N}$, a finite family $\mathcal{F}_{K,e}$ of K -graphs such that

$$\begin{aligned} e \in \mathcal{N} &\implies \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) = T, \\ e \in \mathcal{Y} &\implies \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) > T, \end{aligned}$$

where $\mathcal{Y}$ and $\mathcal{N}$ are recursively inseparable. For every input, the construction also provides an explicit deficient palette together with a specified weighting at which its density polynomial equals T . Hence all instances satisfy the promise in Theorem 1.1.

The fixed-family formulation follows from the same construction.

Theorem 1.2. *For the same K_* and every integer $K \geq K_*$, one can effectively construct a finite family $\mathcal{G}_K$ of K -graphs such that, subject to the promise*

$$\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) \geq \pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K),$$

each of the relations

$$\begin{aligned} \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) = \pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K), \quad \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) \leq \pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K), \\ \text{and} \quad \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}) > \pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K) \end{aligned}$$

is undecidable for finite input families $\mathcal{F}$.

For ordinary Turán density, Pikhurko [17, Question 29] asked whether one can decide $\pi(\mathcal{F}) \leq \alpha$ when the input is a finite family $\mathcal{F}$ of k -graphs and a rational number α . Our theorems give a negative answer to the corresponding comparison problem for uniform Turán density in every sufficiently large uniformity. Here the threshold is fixed in advance, and the reduction still works under the promise that the input density is at least that threshold. We do not address Pikhurko's question for ordinary Turán density. To our knowledge, no undecidability result for exact comparison of uniform Turán densities was previously known.

Undecidability has appeared elsewhere in extremal combinatorics. Hatami and Norine [7] proved that validity of linear inequalities in graph homomorphism densities is undecidable. On the group-theoretic side, Slobodskoi [22] proved that the universal theory of finite groups is undecidable. Our problem differs in both input and output. The input is a finite forbidden hypergraph family, and the output is an exact comparison of uniform Turán densities.

The proof has four layers.

First, we use a consequence of Kharlampovich, Myasnikov, and Sapir [8, Theorem 5.1 and its proof]. There is a fixed finitely presented group and an effective sequence of words $w(e)$ such that, on one recursively inseparable set, $w(e)$ survives in a finite quotient, while on the other it is trivial in every periodic image.

Second, we encode finite permutation actions of this group by partial matchings. Auxiliary reserve graphs give each matching package a fixed quadratic upper bound. If the tested word is trivial in every finite action, a 4-ary relation of positive weighted mass can occur only near the boundary of a partial action. A discrete energy estimate bounds its weighted mass by the total quadratic defect. If the word survives in a finite quotient, the right regular action of that quotient gives a zero-defect construction with positive 4-ary mass.

Third, we translate the resulting finite list of typed relational constraints into ordinary forbidden K -graphs. Two points require care. A target palette may choose its own vertex order, so an ordered query cannot be used directly as an ordinary forbidden hypergraph. An auxiliary ordered design finds either the intended query or its reverse in every vertex order. We then encode all relations by tagged profiles in one 14-coordinate factor. Certification coordinates and pinning records force a deficient target palette to respect the intended type assignment.

Finally, an entropy argument chooses the numbers of certification coordinates so that a prescribed rational type-mass vector is the unique maximizer of the upper bound in the negative case. The choice is effective. An existence argument shows that the search eventually succeeds, and quantifier elimination over the real numbers checks each candidate. Pairing every record with its coordinate reverse deals with the reverse-order case in the explicit constructions. Inserting an unrestricted middle block gives every larger uniformity while preserving both the upper bound and the explicit witnesses.

We prove the weighted blowup step that passes from ordinary palette density to the palette Lagrangian used below. Beyond the Kharlampovich–Myasnikov–Sapir finite-quotient source and the Lin–Sun–Wang–Zhou palette characterization, the proof uses the Erdős–Szekeres theorem and effective quantifier elimination over real closed fields.

The paper is organized as follows. [Section 2](#) introduces uniform density, palettes, and the computability conventions. [Section 3](#) states the group-theoretic source, and [Section 4](#) derives the quantitative finite-action estimate. [Section 5](#) compiles the typed constraints into ordinary finite forbidden families. [Section 6](#) proves the density bound and chooses the rational threshold. Finally, [Section 7](#) constructs the forbidden families and completes the reduction.

2 Uniform Turán density, palettes, and effectiveness

We first fix the combinatorial language used throughout. All hypergraphs, palettes, relations, and type systems are finite unless stated otherwise, and $\mathbb{N} = \{0, 1, 2, \dots\}$. A k -graph is a pair $H = (V(H), E(H))$ with $E(H) \subseteq \binom{V(H)}{k}$. For a positive integer m , write $[m] := \{1, \dots, m\}$. All suprema below are taken in $[0, 1]$; in particular, the supremum of an empty admissible class is 0.

Uniform density. Let $k \geq 3$. For a $(k-2)$ -graph G on a vertex set V , write $\mathcal{K}_k(G)$ for the collection of k -subsets of V that span a complete $(k-2)$ -graph in G . A k -graph H on n vertices is $(d, \mu, k-2)$ -dense if

$$|E(H) \cap \mathcal{K}_k(G)| \geq d |\mathcal{K}_k(G)| - \mu n^k$$

for every $(k-2)$ -graph G on $V(H)$.

For a family $\mathcal{F}$ of k -graphs, its $(k-2)$ -uniform Turán density $\pi_{k-2}^{\text{unif},(k)}(\mathcal{F})$ is the supremum of all $d \in [0, 1]$ such that, for every $\mu > 0$ and every n_0 , there exists an $\mathcal{F}$ -free $(d, \mu, k-2)$ -dense k -graph with at least n_0 vertices.

Palettes. A k -palette is a pair $P = (C(P), A(P))$, where $C(P)$ is a nonempty finite set of colors and $A(P) \subseteq C(P)^k$ is the set of allowed ordered tuples. For a probability vector $\mathbf{x} = (x_c)_{c \in C(P)}$, where $x_c \geq 0$ and $\sum_{c \in C(P)} x_c = 1$, define

$$p_P(\mathbf{x}) := \sum_{(c_1, \dots, c_k) \in A(P)} x_{c_1} \cdots x_{c_k}, \quad \Lambda(P) := \max_{\mathbf{x}} p_P(\mathbf{x}).$$

The maximum exists by compactness of the simplex. The number $\Lambda(P)$ is the *Lagrangian* of P , while $d(P) := |A(P)|/|C(P)|^k$ is its ordinary, unweighted density.

Let F be a k -graph. We say that F is P -colorable if there are a total order $<$ on $V(F)$ and a coloring $\varphi: \binom{V(F)}{k-1} \rightarrow C(P)$ such that, for every edge $e = \{v_1 < \dots < v_k\}$, the tuple $(\varphi(e \setminus \{v_1\}), \dots, \varphi(e \setminus \{v_k\}))$ belongs to $A(P)$. A palette is $\mathcal{F}$ -deficient if no member of $\mathcal{F}$ is P -colorable.

A *palette homomorphism* $f: P \rightarrow Q$ is a map $f: C(P) \rightarrow C(Q)$ such that $(c_1, \dots, c_k) \in A(P)$ implies $(f(c_1), \dots, f(c_k)) \in A(Q)$. If $P \rightarrow Q$ and Q is $\mathcal{F}$ -deficient, then P is $\mathcal{F}$ -deficient.

For an ordered k -graph $(J, <)$, its *canonical palette* $\text{Can}(J, <)$ has as colors the covered $(k-1)$ -faces of J ; each edge $e = \{v_1 < \dots < v_k\}$ contributes the tuple $(e \setminus \{v_1\}, \dots, e \setminus \{v_k\})$. Thus J is P -colorable in the fixed order $<$ if and only if $\text{Can}(J, <) \rightarrow P$. When the order is fixed by the context, we abbreviate $\text{Can}(J, <)$ to $\text{Can}(J)$.

The *reverse palette* $\text{rev } P$ has the same color set and $A(\text{rev } P) := \{(c_k, \dots, c_1) : (c_1, \dots, c_k) \in A(P)\}$. Likewise, $\text{rev } \text{Can}(J, <)$ denotes coordinate reversal of the canonical palette.

The following result is the bridge from palettes to uniform Turán density.

Theorem 2.1 (Lin–Sun–Wang–Zhou [14, Theorem 1.4]). *For every $k \geq 3$ and every family $\mathcal{F}$ of k -graphs,*

$$\pi_{k-2}^{\text{unif},(k)}(\mathcal{F}) = \sup\{d(P) : P \text{ is a finite } \mathcal{F}\text{-deficient } k\text{-palette}\}.$$

We use the following weighted form. It is a blowup consequence of Theorem 2.1; we include the short argument because the weighting will be part of the reduction.

Proposition 2.2. *For every $k \geq 3$ and every family $\mathcal{F}$ of k -graphs,*

$$\pi_{k-2}^{\text{unif},(k)}(\mathcal{F}) = \sup\{\Lambda(P) : P \text{ is a finite } \mathcal{F}\text{-deficient } k\text{-palette}\}.$$

Proof. For every palette P , the uniform distribution on $C(P)$ gives $\Lambda(P) \geq d(P)$. Hence the right-hand side is at least $\pi_{k-2}^{\text{unif},(k)}(\mathcal{F})$ by Theorem 2.1.

For the reverse inequality, fix an $\mathcal{F}$ -deficient palette P and a probability vector $\mathbf{x}$ on $C(P)$. Replace each color c by N_c clones, where $N := \sum_c N_c$ and $N_c/N \rightarrow x_c$, and pull back every allowed tuple of P to all tuples of clones. Projection to the original colors is a palette homomorphism, so every clone palette is still $\mathcal{F}$ -deficient. Its unweighted density tends to $p_P(\mathbf{x})$. By Theorem 2.1, each of these unweighted densities is at most $\pi_{k-2}^{\text{unif},(k)}(\mathcal{F})$; taking the limit gives $p_P(\mathbf{x}) \leq \pi_{k-2}^{\text{unif},(k)}(\mathcal{F})$. Now maximize over $\mathbf{x}$ and take the supremum over P . $\square$

The upper-bound argument later uses loopless reserve graphs. The same cloning argument lets us work with palettes whose allowed tuples contain no repeated color.

Lemma 2.3. *Let P be a finite k -palette and let $\mathbf{x}$ be a probability vector on $C(P)$. There are finite palettes P_N and homomorphisms $P_N \rightarrow P$ such that*

- (a) *every allowed tuple of P_N has pairwise distinct colors; and*
- (b) *$d(P_N) \rightarrow p_P(\mathbf{x})$.*

Consequently, if P is $\mathcal{F}$ -deficient, then every P_N is $\mathcal{F}$ -deficient.

Proof. Choose clone numbers N_c with $N := \sum_c N_c$ and $N_c/N \rightarrow x_c$, and retain only clone tuples with pairwise distinct entries. If a color c occurs q times in an allowed tuple of P , its contribution N_c^q is replaced by the falling factorial $(N_c)_q$. For every c with $x_c > 0$, one has $(N_c)_q/N_c^q \rightarrow 1$, while every tuple containing a color of weight zero has limiting contribution zero and that color may be omitted. Summing over the finitely many allowed tuples shows that $d(P_N) \rightarrow p_P(\mathbf{x})$. Projection to P is a homomorphism, which proves the final assertion. $\square$

A palette satisfying the first conclusion of Lemma 2.3 will be called *repetition-free*. This reduction is important because the reserve relations below are loopless, while loops cannot be excluded directly by an ordinary forbidden hypergraph.

Computability. All effectiveness statements use the ordinary Turing model. A finite hypergraph is encoded by a vertex number and an edge list, a finite family by a finite list of such encodings, and a word by a finite string over the fixed generator alphabet. Finite relations, palettes, ordered hypergraphs, and type words are encoded by their finite tables. Thus *effectively constructible* means that a total algorithm outputs the relevant finite table or incidence list.

The existence arguments used below are converted into terminating algorithms. The monotone block design is found by exhaustive enumeration followed by a finite verification over all vertex orders, and the calibration parameter is found by enumerating the allowed integers and applying quantifier elimination over real closed fields. Once the fixed presentation is chosen, the lower bound K_* and the threshold T are fixed outputs of these searches. For every $K \geq K_*$, the comparison family $\mathcal{G}_K$ and the input family $\mathcal{F}_{K,e}$ are then obtained by finite enumeration.

The value of K_* obtained by the proof is computable but very large. The theorem applies to every $K \geq K_*$, but no claim is made here for the specific uniformity 3.

3 The fixed finite-quotient source

This section states the group-theoretic source in the form used by the reduction. We first recall the relevant terms for readers from combinatorics. A *finite presentation* $\langle S \mid \mathcal{R} \rangle$ specifies a group by generators S and relators $\mathcal{R}$. A *finite quotient* is a finite homomorphic image of that group. A group is *periodic* if every element has finite order; in particular, every finite group and every finite permutation group is periodic.

Two disjoint sets $\mathcal{Y}, \mathcal{N} \subseteq \mathbb{N}$ are *recursively inseparable* if there is no recursive set $A \subseteq \mathbb{N}$ such that $\mathcal{Y} \subseteq A$ and $A \cap \mathcal{N} = \emptyset$.

The required presentation is a consequence of the construction of Kharlampovich, Myasnikov, and Sapir.

Theorem 3.1 (Consequence of Kharlampovich–Myasnikov–Sapir [8, Theorem 5.1 and its proof]). *There exist a fixed nonempty finite generating set S_0 , a fixed finite group presentation $\Gamma := \langle S_0 \mid \mathcal{R} \rangle$, an effective sequence of words $w(e)$ over $S_0^{\pm 1}$, and recursively inseparable sets $\mathcal{Y}, \mathcal{N} \subseteq \mathbb{N}$ such that*

$$\begin{aligned} e \in \mathcal{Y} &\implies \text{some finite quotient } H \text{ of } \Gamma \text{ satisfies } w(e) \neq 1, \\ e \in \mathcal{N} &\implies \text{every homomorphism from } \Gamma \text{ to a periodic group kills } w(e). \end{aligned}$$

In particular, in the second case $w(e)$ is trivial in every finite quotient and every finite permutation image of Γ .

Proof. In the proof of the cited theorem, Kharlampovich, Myasnikov, and Sapir fix a Minsky machine, that is, a counter machine encoding a computation. The resulting group $G'(MM_4)$ has a fixed finite generating set S_0 and a fixed finite system $\mathcal{R}$ of group equations, obtained by writing out the finite relation families (G1)–(G8) with the BR-conjoint conditions expanded. They define two sets $X, X' \subseteq \mathbb{N}$: the machine accepts the input configuration indexed by $e \in X$, whereas its computation from the configuration indexed by $e \in X'$ is infinite. Their construction makes X and X' recursively inseparable.

In that construction, the finite antecedent $\bigwedge \mathcal{R}$ is fixed, whereas the input e effectively determines the finite conclusion group term in the resulting universal formula. We define $w(e)$ to be the ordinary word obtained by recursively expanding this term. The operation $*$ and its iterates are expanded using (G8), and any remaining notation $x(\vec{i}, u)$ is expanded by the finite recursion in the proof of [8, Lemma 4.11]. Since the conclusion term is finite and all these rules are explicit, this is a total effective procedure over the fixed alphabet $S_0^{\pm 1}$.

The cited proof establishes the following two facts for this same conclusion term, and hence for its expansion $w(e)$. If $e \in X$, then the universal implication $\bigwedge \mathcal{R} \implies w(e) = 1$ fails under some assignment in a finite group. If $e \in X'$, then this implication holds under every assignment in every periodic group.

Set $\mathcal{Y} := X$ and $\mathcal{N} := X'$. Replace each equation $u = v$ in $\mathcal{R}$ by the relator uv^{-1} , and let Γ be the resulting fixed presentation. An assignment satisfying the original equations in a group L is exactly a homomorphism $\Gamma \rightarrow L$. When $e \in \mathcal{Y}$, restrict the finite target of a falsifying assignment to the image of this homomorphism. This gives a finite quotient in which $w(e)$ survives. When $e \in \mathcal{N}$, every homomorphism from Γ to a periodic group kills $w(e)$. All expansions are effective, and neither S_0 nor $\mathcal{R}$ depends on e . $\square$

The choice $\mathcal{N} = X'$ is essential: X' is the nonhalting set defined in the proof of [8, Theorem 5.1], not necessarily the whole complement of the accepting set. The periodic-image argument uses the fact that the corresponding computation is infinite.

We keep S_0 as a set of positive generators. For each $s \in S_0$ the construction below uses one pair of links; an occurrence of s^{-1} traverses those same links in reverse. Thus there is no second link pair indexed by the formal inverse of s . We delete empty relators from $\mathcal{R}$, which does not change the presented group, and hence every written relator below is nonempty. Fix $s_0 \in S_0$. For the remainder of the proof we use the written word

$$w^\dagger(e) := w(e)s_0s_0^{-1}, \quad (3.1)$$

without freely reducing the final detour. It represents the same group element, but after the two-link expansion used below its path has length at least four. This prevents the word query from degenerating into two distinct records sharing both of their active colors.

4 A weighted finite-action construction

This section turns the finite-quotient dichotomy into a weighted relational construction. We first define a quadratic package score and the finite list of forbidden local patterns. We then clean the positive support, prove a boundary estimate, and construct exact witnesses for the two sides of the dichotomy.

Fix $e \in \mathbb{N}$ throughout this section and the next two sections, and abbreviate $w^\dagger(e)$ to $w^\dagger$.

We use four copies of the same partial action; these copies will be called *rails*. For every type Z we have a finite set, also denoted by Z , whose elements carry nonnegative weights $(\omega_z)_{z \in Z}$. The type sets are pairwise disjoint, so each element has a unique type. For any subset W of their disjoint union, write

$$a_W := \sum_{z \in W} \omega_z, \quad p_2(W) := \sum_{z \in W} \omega_z^2, \quad e_2(W) := \sum_{\{z, z'\} \in \binom{W}{2}} \omega_z \omega_{z'}.$$

The type masses form the vector $\mathbf{a} = (a_Z)$. Thus

$$p_2(Z) + 2e_2(Z) = a_Z^2. \quad (4.1)$$

No normalization is needed in this section, although in the palette application the total weight over all types is one.

4.1 The package score

For each $r \in [4]$ introduce a master type X_r . For every generator $s \in S_0$, introduce an auxiliary type $Y_{r,s}$ and two bipartite relations $A_{r,s}, B_{r,s} \subseteq X_r \times Y_{r,s}$. The two relations have the same *stored* coordinate order, namely $X_r, Y_{r,s}$. Their path directions differ. An $A_{r,s}$ -edge is traversed from its first entry to its second, whereas a $B_{r,s}$ -edge is traversed from its second entry to its first. Thus, when both relations are bijections, the two-link path

$$X_r \xrightarrow{A_{r,s}} Y_{r,s} \xrightarrow{B_{r,s}} X_r$$

induces the permutation $A_{r,s}B_{r,s}$ of X_r , with the factors written in path order. A letter s^{-1} is traversed by using the same two links in the reverse order and direction. Throughout the action argument, permutations act on the right and products are read in path order. Thus $x(\sigma\tau) = (x\sigma)\tau$. The permutation associated with a written word is obtained by traversing its letters from left to right. We use $\text{Sym}(X)$ with this right-action convention.

Let $\mathcal{D}$ be the fixed collection of these link relations. If $D \in \mathcal{D}$, write U_D, V_D for its first and second stored types, respectively, and attach loopless reserve graphs $N_{D,U_D} \subseteq \binom{U_D}{2}$ and $N_{D,V_D} \subseteq \binom{V_D}{2}$. Put

$$\begin{aligned} h(D) &:= \sum_{(u,v) \in D} \omega_u \omega_v, \\ n(N) &:= \sum_{\{z,z'\} \in N} \omega_z \omega_{z'}, \\ e_D &:= h(D) + n(N_{D,U_D}) + n(N_{D,V_D}). \end{aligned}$$

For any type-indexed vector $\mathbf{t} = (t_Z)$, set $b_D(\mathbf{t}) := \frac{1}{2}(t_{U_D}^2 + t_{V_D}^2)$. We call the link together with its two reserve graphs a *package*; e_D is its *score* and $b_D(\mathbf{a})$ is its *target value*.

Finally introduce a 4-ary relation

$$M \subseteq X_1 \times X_2 \times X_3 \times X_4, \quad m(M) := \sum_{(x_1, x_2, x_3, x_4) \in M} \prod_{r=1}^4 \omega_{x_r}.$$

The next elementary inequality bounds the link part of a package. It will be combined with the reserve graphs to obtain the target value $b_D(\mathbf{a})$.

For a bipartite relation $D \subseteq U \times V$, let $\text{Coll}_U(D)$ be the *collision graph* on U in which distinct x, x' are adjacent when they have a common D -neighbor. Define $\text{Coll}_V(D)$ analogously and put

$$c_U(D) := \sum_{\{x, x'\} \in \text{Coll}_U(D)} \omega_x \omega_{x'}, \quad c_V(D) := \sum_{\{y, y'\} \in \text{Coll}_V(D)} \omega_y \omega_{y'}.$$

Lemma 4.1. *For every finite weighted bipartite relation $D \subseteq U \times V$,*

$$h(D) \leq \frac{1}{2}(p_2(U) + p_2(V)) + c_U(D) + c_V(D).$$

Proof. Let Φ be the left-hand side minus the right-hand side of the claimed inequality. Suppose that $x, x' \in U$ form a collision pair and both have positive weight. Keep $\omega_x + \omega_{x'}$ fixed and transfer weight between x and x' . The contribution

$$-\frac{1}{2}(\omega_x^2 + \omega_{x'}^2) - \omega_x \omega_{x'} = -\frac{1}{2}(\omega_x + \omega_{x'})^2$$

from $p_2(U)$ and the collision edge xx' is constant, while every other term of Φ is affine in the transfer parameter. We may therefore move all weight to one endpoint without decreasing Φ . The same argument applies to a positive-weight collision pair in V .

Each transfer strictly decreases the positive support. Iteration therefore ends with a weighting for which both collision graphs are independent on that support, so the restriction of D is a matching. Consequently

$$h(D) = \sum_{uv \in D} \omega_u \omega_v \leq \frac{1}{2} \sum_{uv \in D} (\omega_u^2 + \omega_v^2) \leq \frac{1}{2}(p_2(U) + p_2(V)),$$

and the two collision sums vanish. Thus the final value of Φ is at most zero. Since the transfers did not decrease Φ , its original value is also at most zero. $\square$

In any structure avoiding the map-collision constraints defined below, every collision pair of D is absent not only from the reserve attached to D , but from *every* reserve whose vertex type is the relevant type. Thus, for $D \subseteq U_D \times V_D$,

$$n(N_{D, U_D}) \leq e_2(U_D) - c_{U_D}(D), \quad n(N_{D, V_D}) \leq e_2(V_D) - c_{V_D}(D).$$

Combining these inequalities with [Lemma 4.1](#) and [\(4.1\)](#) gives the pointwise package bound

$$e_D \leq b_D(\mathbf{a}). \tag{4.2}$$

4.2 The finite list of local constraints

We now formalize the local patterns that must be excluded and what it means for such a pattern to occur. Each relation atom will later be represented by one palette record.

Definition 4.2. A *typed query* is a finite relational pattern whose variables carry specified types. Its atoms use the ordered binary relation symbols D , the loopless unordered reserve symbols N , and the ordered 4-ary symbol M ; it may also contain explicitly stated disequalities. An *occurrence* of a query in a typed structure is a type-preserving assignment of its variables that satisfies every relation atom and every stated disequality. No injectivity is required beyond those disequalities.

Variables written with the same name are identified. A variable declared *private* occurs in one specified relation atom and nowhere else, unless a later pinning operation explicitly identifies it with a certification tuple. For an unordered reserve atom $N(x, y)$, satisfaction means $\{x, y\} \in N$; in particular, looplessness enforces $x \neq y$. Every disequality needed below is witnessed inside such a reserve atom. This is why, after the repetition-free approximation of Lemma 2.3, palette homomorphisms cannot collapse a required distinct pair.

We make the signature and the path expansion explicit. Its sorts are X_r and $Y_{r,s}$ for $r \in [4]$ and $s \in S_0$. It has the ordered binary symbols $A_{r,s}, B_{r,s}$, the symmetric irreflexive symbols $N_{D,Z}$ for every $D \in \mathcal{D}$ and $Z \in \{U_D, V_D\}$, and the ordered 4-ary symbol M . For a letter s^η , where $\eta \in \{1, -1\}$, define

$$\begin{aligned}\text{Path}_s(x, x'; y) &:= A_{r,s}(x, y) \wedge B_{r,s}(x', y), \\ \text{Path}_{s^{-1}}(x, x'; y) &:= B_{r,s}(x, y) \wedge A_{r,s}(x', y).\end{aligned}$$

Here x, x' have sort X_r and y has sort $Y_{r,s}$. If $z = s_1^{\eta_1} \cdots s_t^{\eta_t}$ is a written word, then $\text{Path}_{z,r}(x_0, x_t)$ denotes the finite pattern with fresh variables $x_1, \dots, x_{t-1}, y_1, \dots, y_t$ and atoms

$$\bigwedge_{i=1}^t \text{Path}_{s_i^{\eta_i}}(x_{i-1}, x_i; y_i),$$

where the x_i have sort X_r and y_i has sort Y_{r,s_i} . Thus $\text{Path}_{z,r}$ is notation for a finite conjunction of $2t$ relation atoms, not a formula with logical quantifiers in the finite relational language.

We now specify the finite typed patterns that are forbidden. The list depends on the fixed presentation $\Gamma = \langle S_0 \mid \mathcal{R} \rangle$ and, only in its last item, on the tested word $w^\dagger$.

(Q1) *Map collisions*. For $D \subseteq U \times V$, and for every reserve symbol N on V , forbid

$$D(x, y), \quad D(x, y'), \quad N(y, y').$$

The reserve atom itself forces $y \neq y'$. For every reserve N on U , also forbid the incoming pattern

$$D(x, y), \quad D(x', y), \quad N(x, x').$$

The reserve may have a package tag unrelated to D . This cross-expansion is needed in the common support reduction.

(Q2) *Relators*. For $q \in \mathcal{R}$, $r \in [4]$, and every reserve N on X_r , forbid

$$\text{Path}_{q,r}(x, x') \quad \text{together with} \quad N(x, x').$$

(Q3) *Cyclic functionality of M* . Indices are read modulo four. For $r \in [4]$ and every reserve N on X_{r+1} , forbid

$$M(x_1, \dots, x_r, \dots, x_4), \quad M(x'_1, \dots, x_r, \dots, x'_4), \quad N(x_{r+1}, x'_{r+1}).$$

(Q4) *The word test*. For each $r \in [4]$, forbid

$$M(x_1, x_2, x_3, x_4) \quad \text{together with} \quad \text{Path}_{w^\dagger, r}(x_r, x_r).$$

The reserve graphs are loopless, so a relator path that closes is compatible with (Q2). Likewise, the word test records actual equality of the two endpoints of the expanded path.

4.3 Cleaning the positive support

Suppose throughout this subsection that the finite weighted typed structure avoids queries (Q1)–(Q3). For each type Z , form a conflict graph C_Z whose edge set consists of the following pairs.

- all map-collision pairs lying in Z ;
- all distinct endpoint pairs of fully defined, nonclosing expanded relator paths based at $Z = X_r$; and
- all pairs in coordinate $r + 1$ arising from two M -tuples that agree in coordinate r .

By the cross-expanded constraints above, every reserve on Z omits every edge of C_Z .

The next lemma replaces an arbitrary weighting by one whose positive support behaves like a partial group action. This is where the cross-expanded reserve constraints are used.

Lemma 4.3. *Suppose the typed structure avoids (Q1)–(Q3). Fix the total mass in every type. Let Ψ be any nonnegative linear combination of the package scores e_D and the 4-ary score $m(M)$. There is a redistribution of weight within the types, followed by adding reserve edges, that does not decrease Ψ and has the following properties on its positive support.*

- (a) *every $D \in \mathcal{D}$ is a partial matching;*
- (b) *every fully defined relator path closes;*
- (c) *M is a partial four-matching, meaning that each coordinate projection $M \rightarrow X_r$ is injective; and*
- (d) *every reserve is complete on the positive support of its type.*

For the resulting structure,

$$\Delta := \sum_{D \in \mathcal{D}} (b_D(\mathbf{a}) - e_D) = \sum_{D \in \mathcal{D}} \left[\frac{1}{2} (p_2(U_D) + p_2(V_D)) - h(D) \right].$$

The repaired structure still avoids (Q1)–(Q3); if it initially avoids (Q4), then it still does so after the repair.

Proof. Let zz' be an edge of a conflict graph on a type Z . Every monomial in a link score or in $m(M)$ contains at most one element of Z . Moreover, every reserve on Z omits zz' , so no reserve score contains the product $\omega_z \omega_{z'}$. Consequently, with $\omega_z + \omega_{z'}$ fixed, the objective Ψ is affine in ω_z . Move all mass to an endpoint at which Ψ does not decrease. Each move decreases the positive support, and iteration produces a support independent in every conflict graph.

Map collisions are then absent on the positive support, proving (a). If a fully defined relator path had distinct positive endpoints, those endpoints would be adjacent in the corresponding conflict graph, proving (b). For (c), if two positive M -tuples agree in coordinate r , their coordinate- $(r+1)$ entries cannot be a conflict pair, and hence must agree. Going cyclically around the four coordinates shows that the two tuples are equal.

Finally add every missing pair of distinct positive vertices to each reserve. No forbidden query is created, because every pair capable of completing one of (Q1)–(Q3) was placed in the relevant conflict graph. This only increases Ψ , proving (d). Thus the repair preserves avoidance of (Q1)–(Q3). It also preserves avoidance of (Q4), if that query was absent initially, because the word query has no reserve atom and the repair changes no nonreserve relation. With complete reserves, (4.1) gives $b_D(\mathbf{a}) - e_D = \frac{1}{2} (p_2(U_D) + p_2(V_D)) - h(D)$, which proves the displayed identity. $\square$

After applying Lemma 4.3, perform the following *positive-support reduction*. Replace each type Z by $Z^+ := \{z \in Z : \omega_z > 0\}$ and every relation by its restriction to the corresponding positive-support types. In particular, an edge with a zero-weight endpoint is regarded as absent. This operation changes neither the type masses nor any of the quantities p_2 , $h(D)$, $n(N)$, $m(M)$, and Δ , because every discarded monomial contains a zero factor. Properties (a)–(d) of Lemma 4.3 therefore hold literally on the resulting finite structure. All matching graphs and components in the boundary argument below refer to this positive-support structure.

4.4 The boundary estimate

Fix a rail r and work in the positive-support structure just defined. Regard the restrictions of $A_{r,s}$ and $B_{r,s}$ as labeled partial matching edges. For each relation symbol and each incident side, there is one required labeled half-edge. Thus a master vertex has at most $2|S_0|$ required half-edges, and an auxiliary vertex has at most two. Set $\kappa := 2|S_0|$. A connected component of the present matching graph is called *boundaryless* if every required labeled half-edge at each of its vertices is present; otherwise it is a *boundary component*.

For every missing required half-edge at a vertex x , attach a distinct new cemetery vertex of weight zero and the corresponding labeled edge. For a fixed partial matching $D \subseteq U^+ \times V^+$, write $\tilde{D}$ for the completed matching. This completion makes every positive vertex in $U^+ \cup V^+$ incident with exactly one

D -edge, present or cemetery. Consequently

$$\begin{aligned} \sum_{xy \in \tilde{D}} (\omega_x - \omega_y)^2 &= p_2(U^+) + p_2(V^+) - 2h(D) \\ &= 2 \left[\frac{1}{2} (p_2(U^+) + p_2(V^+)) - h(D) \right]. \end{aligned} \quad (4.3)$$

For every component C on rail r , define its defect contribution by

$$\Delta_C := \sum_{\substack{D \in \mathcal{D} \\ D \text{ on rail } r}} \left[\frac{1}{2} \sum_{z \in C \cap (U_D \cup V_D)} \omega_z^2 - \sum_{\substack{xy \in D \\ x, y \in C}} \omega_x \omega_y \right].$$

Applying (4.3) componentwise shows that $\Delta_C \geq 0$, and the matching energies decompose over the rail components, so

$$\Delta = \sum_{r=1}^4 \sum_{C \text{ on rail } r} \Delta_C.$$

If C is a boundary component, write $\tilde{E}_C$ for the labeled multiset of its present edges together with all labeled cemetery edges incident with vertices of C . Summing (4.3) over the relation labels and restricting the resulting labeled edge sum to C gives the exact identity

$$\sum_{xy \in \tilde{E}_C} (\omega_x - \omega_y)^2 = 2\Delta_C.$$

The following fourth-moment estimate is the quantitative part of the boundary argument.

Lemma 4.4. *For every boundary component C , with $\mu_C := \sum_{x \in C} \omega_x$, one has*

$$\sum_{x \in C} \omega_x^4 \leq 4\kappa \mu_C^2 \Delta_C.$$

Proof. Because C is connected and has a missing required half-edge, every $x \in C$ is joined in $\tilde{E}_C$ to a cemetery vertex. Choose such a path $x = x_0, x_1, \dots, x_t$, where $\omega_{x_t} = 0$. Then

$$\omega_x^2 = \sum_{i=0}^{t-1} (\omega_{x_i}^2 - \omega_{x_{i+1}}^2) = \sum_{i=0}^{t-1} (\omega_{x_i} - \omega_{x_{i+1}})(\omega_{x_i} + \omega_{x_{i+1}}).$$

Cauchy-Schwarz, followed by enlarging both sums from the path to $\tilde{E}_C$, gives

$$\begin{aligned} \omega_x^4 &\leq \left(\sum_{yz \in \tilde{E}_C} (\omega_y - \omega_z)^2 \right) \left(\sum_{yz \in \tilde{E}_C} (\omega_y + \omega_z)^2 \right) \\ &\leq (2\Delta_C)(2\kappa p_2(C)) = 4\kappa \Delta_C p_2(C). \end{aligned}$$

Here $p_2(C) = \sum_{x \in C} \omega_x^2$. Every noncemetery vertex is incident with at most κ completed labeled edges and every cemetery vertex has weight zero. Hence

$$\sum_{yz \in \tilde{E}_C} (\omega_y + \omega_z)^2 \leq 2\kappa p_2(C).$$

Since $p_2(C) \leq \mu_C \max_{x \in C} \omega_x$, applying the preceding inequality at a maximum-weight vertex yields $(\max_{x \in C} \omega_x)^3 \leq 4\kappa \mu_C \Delta_C$. Therefore

$$\sum_{x \in C} \omega_x^4 \leq \mu_C \left(\max_{x \in C} \omega_x \right)^3 \leq 4\kappa \mu_C^2 \Delta_C. \quad \square$$

For later use, note that a component with no boundary is an actual finite action of the presented group. Indeed, let C be a boundaryless component on rail r , and put $X_C := C \cap X_r$. Every auxiliary vertex is incident only with master vertices, so X_C is nonempty. For each s , completeness of the required half-edges and the partial-matching property show that $A_{r,s}$ and $B_{r,s}$ restrict to perfect matchings between X_C and $Y_{r,s} \cap C$. Their two-link composition is therefore a permutation of X_C . Every expanded relator path is fully defined inside C and closes by [Lemma 4.3](#). Hence the generator permutations satisfy all relations in $\mathcal{R}$ and define a homomorphism $\rho_C: \Gamma \rightarrow \text{Sym}(X_C)$. For $s \in S_0$, the permutation $\rho_C(s)$ follows the unique $A_{r,s}$ -edge from X_C to $Y_{r,s} \cap C$ and then the unique $B_{r,s}$ -edge back to X_C .

Say that a written word z has *property (W)* if

$$\rho(z) = \text{id}_X \quad \text{for every finite set } X \text{ and every homomorphism } \rho: \Gamma \rightarrow \text{Sym}(X).$$

If $e \in \mathcal{N}$, then $w^\dagger$ has property (W) by [Theorem 3.1](#). Whenever the tested word has property (W), it acts trivially on every boundaryless component. Query (Q4) then prevents a coordinate of a positive-weight M -tuple from lying in such a component. This observation and [Lemma 4.4](#) give the inequality needed in the negative case.

Theorem 4.5. *Assume that the tested word has property (W), the total weight over all types is at most one, and the typed structure avoids (Q1)–(Q4). Then*

$$m(M) \leq \kappa \sum_{D \in \mathcal{D}} (b_D(\mathbf{a}) - e_D).$$

Proof. First apply [Lemma 4.3](#) to the objective $\Psi := \sum_{D \in \mathcal{D}} e_D + \kappa^{-1} m(M)$. Then pass to the positive-support structure described above; all terms of the objective and all type masses are unchanged. For this repaired structure, every coordinate of a tuple in M lies in a boundary component and M is a partial four-matching. For each $(x_1, x_2, x_3, x_4) \in M$, the arithmetic–geometric mean inequality gives

$$\prod_{r=1}^4 \omega_{x_r} \leq \frac{1}{4} \sum_{r=1}^4 \omega_{x_r}^4.$$

Each coordinate projection of M is injective, so summing this inequality and applying [Lemma 4.4](#) gives

$$\begin{aligned} m(M) &\leq \frac{1}{4} \sum_{r=1}^4 \sum_{\substack{C \text{ boundary} \\ \text{on rail } r}} \sum_{x \in C \cap X_r} \omega_x^4 \\ &\leq \kappa \sum_{r=1}^4 \sum_{\substack{C \text{ boundary} \\ \text{on rail } r}} \mu_C^2 \Delta_C \leq \kappa \Delta. \end{aligned}$$

For the last inequality, every component mass satisfies $\mu_C \leq \sum_z \omega_z \leq 1$.

For the repaired structure,

$$\sum_D e_D + \kappa^{-1} m(M) = \sum_D b_D(\mathbf{a}) - \Delta + \kappa^{-1} m(M) \leq \sum_D b_D(\mathbf{a}).$$

The repair preserved every type mass and did not decrease the chosen objective. Hence the original structure satisfies the same upper bound, which rearranges to the claimed inequality. $\square$

If the total mass were A rather than at most one, the homogeneous form of [Theorem 4.5](#) would be $m(M) \leq \kappa A^2 \sum_{D \in \mathcal{D}} (b_D(\mathbf{a}) - e_D)$. We use only the normalized form.

4.5 Exact witnesses

We need two constructions that attain all package bounds. The first works for every tested word.

Lemma 4.6. *Suppose all action types joined by a link have a common total mass a . There is a one-point realization of every action type in which every link is the unique matching, every reserve is empty, and $M = \emptyset$. It avoids all four query types for every word and satisfies $e_D = b_D(\mathbf{a})$ for every $D \in \mathcal{D}$.*

Proof. Each link contributes a^2 and its two loopless reserves contribute zero, while $b_D(\mathbf{a}) = a^2$. Map collisions do not occur; every relator path closes but a closing reserve loop is unavailable; and the two query types involving M are absent because M is empty. $\square$

When the tested word survives in a finite quotient, the right regular action gives the second construction. Here *right regular action* means that H acts on itself by right multiplication.

Lemma 4.7. *Suppose $e \in \mathcal{Y}$, and let H be a finite quotient of Γ in which $w^\dagger \neq 1$. If all action types have a common total mass $a > 0$, there is a realization avoiding all local queries such that $e_D = b_D(\mathbf{a})$ for every $D \in \mathcal{D}$ and $m(M) > 0$.*

Proof. On each rail identify every action type with a copy of H , assigning weight $a/|H|$ to each element. Use the right regular action. More explicitly, write $x_h \in X_r$ and $y_h \in Y_{r,s}$ and put

$$A_{r,s} := \{(x_h, y_h) : h \in H\}, \quad B_{r,s} := \{(x_{hs}, y_h) : h \in H\}.$$

Thus the two-link path sends x_h to x_{hs} . Complete every reserve graph on its type. A package contains one matching of weight $a^2/|H|$ and two complete reserve graphs of total weight $a^2(1 - |H|^{-1})$, so its score is $a^2 = b_D(\mathbf{a})$.

Set $M := \{(x_h, x_h, x_h, x_h) : h \in H\}$, using the corresponding copy of h on each rail. Then $m(M) = a^4/|H|^3 > 0$. The links are matchings, all relators close, and M is a four-matching, so the first three query types are absent. Right multiplication by the nonidentity element $w^\dagger$ has no fixed point on H , and hence no word-test query occurs. $\square$

5 Compiling typed constraints into ordinary forbidden hypergraphs

The compiler has three stages. We first remove the freedom to reorder the vertices of a forbidden hypergraph. We then encode all relation symbols inside one fixed coordinate factor. Finally, certification coordinates enforce the intended color types, and canonical incidence constructions turn the resulting palette patterns into ordinary forbidden hypergraphs. Every stage is finite and effective.

We use the following terminology throughout this section. A *record* is an allowed tuple of a palette. A *coordinate factor* is a specified consecutive block of coordinates in a longer record, and its *factor projection* is the projection of all allowed full records onto that block. Given a type assignment, a *type box* is the Cartesian set of tuples having one prescribed type in each coordinate.

5.1 Removing the ambient vertex order

The finite-action construction produces typed, ordered relational constraints. We first remove the freedom to reorder the vertices. A canonical query is naturally an ordered hypergraph, whereas palette colorability of an ordinary hypergraph may use an arbitrary vertex order. The following ordered design lets us find the query, possibly with its order reversed, inside every vertex order.

Lemma 5.1. *For every integer $m \geq 2$, there is an effectively constructible finite ordered linear m -graph D_m with a reference order $<$ such that every total order $\prec$ on $V(D_m)$ contains an edge $b_1 < \dots < b_m$ whose vertices are increasing or decreasing in $\prec$.*

Proof. Set $R_m := (m-1)^2 + 1$. Consider $[N]$ with its natural order and another total order $\prec$. By the Erdős-Szekeres monotone subsequence theorem [4], every R_m -subset contains an m -subset that is monotone between the two orders. Double counting pairs consisting of an R_m -set and a monotone m -subset shows that the number of monotone m -subsets is at least $\binom{N}{m} / \binom{R_m}{m}$.

Retain every m -subset independently with probability $C_m(\log N)/N^{m-1}$, where C_m is a sufficiently large constant. For any fixed order $\prec$, a Chernoff bound shows that the probability of retaining fewer than $c_m C_m N \log N$ monotone blocks is at most $\exp(-c'_m C_m N \log N)$ for positive constants c_m, c'_m . Hence

$$\Pr[\text{some order has too few retained monotone blocks}] \leq N! \exp(-c'_m C_m N \log N) = o(1)$$

once C_m is sufficiently large.

If X is the number of unordered pairs of retained blocks meeting in at least two vertices, then

$$\mathbb{E}X = \left(C_m \frac{\log N}{N^{m-1}} \right)^2 \sum_{j=2}^m O_m(N^{2m-j}) = O_m((\log N)^2).$$

Markov's inequality gives $\Pr[X > N] = o(1)$. Thus, with positive probability, both required events hold simultaneously. In that outcome every order has $\Omega_m(N \log N)$ retained monotone blocks and $X \leq N$. Delete one block from each intersecting pair. For sufficiently large N , every order still has a monotone retained block, while the remaining block system is linear. This proves existence.

Effectivity does not depend on extracting numerical bounds from the random argument. Enumerate finite ordered linear m -graphs and check, over their finitely many vertex orders, whether the stated property holds. The existence argument guarantees termination. $\square$

We now replace an ordered query by an ordinary hypergraph. The two alternatives in the conclusion correspond to the two possible directions of the monotone block.

Lemma 5.2. *Let $(J, <)$ be a finite ordered k -graph with $k \geq 3$ and at least one edge. One can effectively construct a finite ordinary k -graph $\mathfrak{D}(J)$ such that, for every finite k -palette Q ,*

$$\mathfrak{D}(J) \text{ is } Q\text{-colorable} \iff \text{Can}(J, <) \rightarrow Q \text{ or } \text{rev Can}(J, <) \rightarrow Q.$$

Proof. Delete isolated vertices of J and put $m := |V(J)|$. Take the design D_m from Lemma 5.1. On each edge $b_1 < \dots < b_m$ of D_m , place an order-preserving copy of $(J, <)$, and let $\mathfrak{D}(J)$ be the union of these copies after forgetting the reference order. Different design blocks meet in at most one vertex. Since $k \geq 3$, their covered $(k-1)$ -shadows are disjoint.

If $\text{Can}(J, <) \rightarrow Q$, use the reference order of D_m and color the covered $(k-1)$ -faces of all copies independently by the chosen homomorphism. Assign an arbitrary color of Q to every remaining $(k-1)$ -face. If $\text{rev Can}(J, <) \rightarrow Q$, use the reverse reference order in the same way. This proves the reverse implication in the claimed equivalence.

Conversely, suppose that $\mathfrak{D}(J)$ has a Q -coloring under some total vertex order $\prec$. By the design property, one block is monotone or antimonotone between $<$ and $\prec$. Restricting the coloring to the copy of J on this block yields respectively a homomorphism $\text{Can}(J, <) \rightarrow Q$ or $\text{rev Can}(J, <) \rightarrow Q$. $\square$

The equivalence, rather than only one implication, is needed later. In the upper bound, the first branch detects an intended typed query in an arbitrary target. In the explicit constructions, reversal invariance excludes both branches.

5.2 Encoding all relations in one factor

A direct Cartesian product with one factor for each package would separate consecutive atoms of a word path into unrelated coordinate pairs. Such a layout is not compatible with the canonical incidence geometry used later. We instead encode every relation as a tagged profile in one fixed coordinate factor.

Types and coordinates. To the action types $X_r, Y_{r,s}$ add fresh filler types F, F_0, F_1 and tag types Tag_M and Tag_D for $D \in \mathcal{D}$, all mutually distinct. Let $\mathcal{T}$ denote the resulting fixed finite type set.

A *relation factor* has length $L := 14$. The four active coordinates of an M -profile are $c_1 := 3$, $c_2 := 6$, $c_3 := 9$, and $c_4 := 12$. The adjacent coordinate pairs assigned to the four rails are $I_1 := \{11, 12\}$, $I_2 := \{3, 4\}$, $I_3 := \{6, 7\}$, and $I_4 := \{9, 10\}$. Consequently, the reserve used for the cyclic dependency $r \rightarrow r+1$ occupies

$$\{c_r, c_r + 1\} \quad (r = 1, 2, 3), \quad \{c_4 - 1, c_4\} \quad (r = 4), \quad (5.1)$$

which is exactly I_{r+1} , with indices modulo four.

Signal type words. Fix $D \in \mathcal{D}$ on rail r , and denote the two coordinates of I_r in increasing order by $i, i + 1$. Define four *signal type words* of length L as follows. In each word, coordinate 1 has type Tag_D , every inactive coordinate has type F , and the active and terminal coordinates are

profile	$(i, i + 1)$	$(13, 14)$
$D^{(0)}$	(U_D, V_D)	(F_0, F_1)
$D^{(1)}$	(U_D, V_D)	(F_1, F_0)
N_{D, U_D}	(U_D, U_D)	(F_0, F_1)
N_{D, V_D}	(V_D, V_D)	(F_0, F_1)

For a type-mass vector $\mathbf{u} = (u_v)_{v \in \mathcal{T}}$, their common filler monomial is $\theta_D(\mathbf{u}) := u_{\text{Tag}_D} u_{F_0} u_{F_1} u_F^9$.

The M -profile has type X_r in coordinate c_r for every $r \in [4]$, type Tag_M in coordinate 1, types F_0, F_1 in coordinates 13, 14, and type F elsewhere. Its filler monomial is $\beta(\mathbf{u}) := u_{\text{Tag}_M} u_{F_0} u_{F_1} u_F^7$. All signal type words are distinct. Let Σ be their finite collection.

Given pairwise disjoint color classes $(C_v)_{v \in \mathcal{T}}$, the intended realization of the relation factor is a relation in C^L , where $C := \bigcup_v C_v$. We define it box by box according to the following rules.

- every box whose type word is not in Σ is completely allowed;
- the two map boxes are restricted by relations $D^{(0)}, D^{(1)} \subseteq C_{U_D} \times C_{V_D}$;
- the two reserve boxes are restricted by ordered supports on $C_{U_D}^2$ and $C_{V_D}^2$; and
- the M -box is restricted by a relation $M \subseteq C_{X_1} \times \dots \times C_{X_4}$.

Membership in a signal box depends only on its displayed active coordinates; all tag and filler coordinates are free within their type classes.

For a relation $R \subseteq C^I$, write

$$d_\mu(R) := \sum_{(c_i)_{i \in I} \in R} \prod_{i \in I} \mu(c_i)$$

for its weighted density. For an arbitrary factor projection R , let R_σ be its restriction to the signal type box $\sigma \in \Sigma$; the *active support* of R_σ is its projection onto the displayed active coordinates. If the color weights have type-mass vector $\mathbf{u}$, product measure gives

$$d_\mu(R_\sigma) \leq (\text{tag-and-filler monomial for } \sigma) d_\mu(\text{proj}_{\text{act}} R_\sigma). \quad (5.2)$$

Indeed, R_σ is contained in the product of its active support and the full tag-and-filler classes. Thus the bounds below apply to projections of arbitrary target palettes; the product-form intended realization defined above is needed only for equality in the explicit witnesses.

Call a factor projection *repetition-free* if every one of its allowed L -tuples has pairwise distinct colors, as supplied by Lemma 2.3. For such a projection, let $\hat{D}$ be the union of the active supports of the two map boxes. Let N_{D, U_D} and N_{D, V_D} be the loopless undirected active supports obtained by forgetting the orientation of the corresponding reserve boxes, and let M be the active support of the M -signal box. By (5.2), the total contribution of the four D -signal boxes is at most

$$2\theta_D(\mathbf{u}) [h(\hat{D}) + n(N_{D, U_D}) + n(N_{D, V_D})]. \quad (5.3)$$

Indeed, an active map pair can occur in at most two map profiles, and an unordered reserve pair in at most its two orientations. Equality holds in the intended constructions, where the two map copies coincide and both orientations of every reserve edge are present.

Expanding the typed queries. We next replace each atom of a typed query by every profile that can represent it.

Definition 5.3. Let $\mathcal{Q}$ be one of the finite typed queries in Section 4.2. Its finite set $\text{Exp}(\mathcal{Q})$ of *signal-profile expansions* is defined record by record as follows.

- An atom $D(x, y)$ is replaced independently by either the $D^{(0)}$ - or $D^{(1)}$ -profile, with x, y placed in its displayed active coordinates.

- (ii) An atom $N_{D,Z}(x, y)$ is replaced by the corresponding reserve profile in either of its two orientations. Since the abstract query list was already cross-expanded over all reserve symbols on the sort Z , the tag $\mathbf{Tag}_D$ here is the tag of that chosen reserve symbol and need not be the tag of an adjacent map atom.
- (iii) An atom $M(x_1, x_2, x_3, x_4)$ is replaced by the unique M -profile, with x_r in coordinate c_r for every $r \in [4]$.
- (iv) Two active slots receive the same color variable if and only if the corresponding typed variables of $\mathcal{Q}$ are identical. Every other slot of every record receives a new variable, different from all variables assigned to other private slots.

Thus an element of $\text{Exp}(\mathcal{Q})$ is a finite palette pattern whose records all have length L , and its shared colors are exactly the variable identifications written in $\mathcal{Q}$. The definition is algorithmic. It is a finite Cartesian product of two-element choices, followed by a finite relabeling of slots.

We say that such an expansion *occurs* in a factor projection if there is a map from its variables to target colors such that every record chosen from a profile σ maps to an L -tuple in R_σ . Equivalently, occurrence is a profile-preserving palette homomorphism into the L -palette defined by the projection. This map need not be injective, so distinct private variables may have the same image. The arguments below use repetition-freeness only to obtain the disequalities witnessed by reserve atoms.

The expansion is useful because avoiding the expanded palette patterns implies avoidance of the original typed queries.

Lemma 5.4. *Suppose a repetition-free relation-factor projection contains none of the expanded signal-profile queries. Then the union supports $\widehat{D}, N_{D,U_D}, N_{D,V_D}$ and M avoid all abstract queries (Q1)–(Q4). In particular, if the tested word has property (W), they satisfy*

$$m(M) \leq \kappa \sum_{D \in \mathcal{D}} \left(b_D(\mathbf{u}) - h(\widehat{D}) - n(N_{D,U_D}) - n(N_{D,V_D}) \right).$$

Proof. Indeed, if an abstract query occurred in the union supports, choosing for each atom a profile and orientation that witnesses membership in the corresponding union would give one of the expanded queries. The inequality is then Theorem 4.5 applied to the extracted typed structure. $\square$

5.3 Type certification and canonical lifting

The profile expansion still presupposes that colors have the intended types. We now enforce these types by finitely many ordinary forbidden hypergraphs and then realize every expanded query as the canonical palette of an ordered hypergraph.

The type shell. Let $k \geq 3$, let $\mathcal{T}$ be a finite type set, and let $A_1, \dots, A_k \subseteq \mathcal{T}$ be nonempty. Define the *rectangular type shell* to be the k -palette

$$B_A := \left(\mathcal{T}, \prod_{i=1}^k A_i \right).$$

For a palette Q , a homomorphism $Q \rightarrow B_A$ is precisely a global type assignment to the colors of Q that is compatible with every coordinate in which each color occurs.

For a set $I \subseteq [k]$, let the *star palette* $\text{Star}(I)$ consist of one central color z and, for each $i \in I$, one tuple in which z occurs in coordinate i and all other colors are private to that tuple.

The next lemma gives a finite obstruction to the failure of a global type assignment.

Lemma 5.5. *There is an effectively computable finite collection $\mathcal{S}_A$ of ordered k -graphs such that, for every k -palette Q ,*

$$(\mathfrak{D}(J) \text{ is not } Q\text{-colorable for every } J \in \mathcal{S}_A) \implies Q \rightarrow B_A.$$

Every member of $\mathcal{S}_A$ is a canonical realization of a star $\text{Star}(I)$ with $|I| \leq |\mathcal{T}|$.

Proof. A map $\vartheta: C(Q) \rightarrow \mathcal{T}$ is a homomorphism to B_A if and only if, for every color $c \in C(Q)$,

$$\vartheta(c) \in \bigcap_{\substack{i: c \text{ occurs in coordinate } i \\ \text{of an allowed tuple of } Q}} A_i.$$

An empty intersection means $\mathcal{T}$. Thus $Q \rightarrow B_A$ if and only if every displayed intersection is nonempty, since the values $\vartheta(c)$ may be chosen independently.

If the displayed intersection is empty, choose an inclusion-minimal subset I of its indexing set such that $\bigcap_{i \in I} A_i = \emptyset$. For every $i \in I$, choose $t_i \in \bigcap_{j \in I \setminus \{i\}} A_j \setminus A_i$. The elements t_i are pairwise distinct, and hence $|I| \leq |\mathcal{T}|$. The corresponding star palette maps to Q by sending its central color to c and choosing a witnessing allowed tuple for each occurrence.

There are only finitely many inclusion-minimal $I \subseteq [k]$ with $\bigcap_{i \in I} A_i = \emptyset$. The star $\text{Star}(I)$ is canonical. Take a common ordered $(k-1)$ -set C for the central color and, for every $i \in I$, add a new outside vertex in the i th gap of C . The resulting edge has C as its coordinate- i face, while all other covered faces are private. Let J_I be the resulting ordered k -graph and put

$$\mathcal{S}_A := \left\{ J_I : I \subseteq [k] \text{ is inclusion-minimal and } \bigcap_{i \in I} A_i = \emptyset \right\}.$$

If the star maps to Q , then $\text{Can}(J_I) \rightarrow Q$, and [Lemma 5.2](#) makes $\mathcal{D}(J_I)$ Q -colorable. This proves the contrapositive of the claimed implication. $\square$

Certification coordinates and pins. We now instantiate the preceding shell construction for a provisional full-record layout. Fix integers $R \geq 1$, $m_v \geq 1$ for $v \in \mathcal{T}$, and $h \geq 0$, and put $K^{\text{prov}} := 2(\sum_{v \in \mathcal{T}} m_v + RL) + h$. Partition the K^{prov} record coordinates into a base half consisting of the certification blocks and R relation-factor copies, an unrestricted middle block of length h , and the coordinate reverse of the base half. In the resulting instance of B_A , a certification coordinate in the block for v has shell set $\{v\}$, whereas every relation-factor and middle coordinate has shell set $\mathcal{T}$. The calibration in [Section 6.2](#) will specialize R and the m_v ; the final construction then takes $h = K - K_*$.

Let Q be a target palette with a type assignment $\vartheta: Q \rightarrow B_A$, and equip its colors with a probability distribution μ . For one fixed half of the record, let C_v^0 be the set of v -typed colors that occur in at least one of the m_v certification coordinates of an allowed tuple of Q , and put

$$\mathbf{u} := (u_v)_{v \in \mathcal{T}}, \quad u_v := \mu(C_v^0). \quad (5.4)$$

The projection of Q to all certification coordinates is contained in $\prod_{v \in \mathcal{T}} (C_v^0)^{m_v}$ and therefore has weighted density at most

$$\prod_{v \in \mathcal{T}} u_v^{m_v}. \quad (5.5)$$

The sets C_v^0 are pairwise disjoint, so $\sum_v u_v \leq 1$.

A signal record in a relation factor is called *fully certified* if all of its L coordinate colors belong to the corresponding sets C_v^0 .

To connect every local profile color to the global type assignment, we add one certification record for each profile coordinate.

Definition 5.6. Fix a typed query $\mathcal{Q}$, an expansion $E \in \text{Exp}(\mathcal{Q})$, and a relation-factor copy in the base half. A *base-half pinned full-record lift* of E is obtained by the following finite procedure.

- (i) Replace every length- L record ρ of E by a full record $\hat{\rho}$. The designated relation factor of $\hat{\rho}$ is ρ , with exactly the variable identifications prescribed by E . Every slot in the other relation factors, in the certification block, in the opposite half, and in the unrestricted middle block receives a fresh private variable.
- (ii) For every pair (ρ, j) consisting of a record of E and a slot $j \in [L]$, let $z_{\rho, j}$ be the variable in that slot and let v be its type. Choose one of the m_v certification coordinates of type v , and add a full record whose chosen coordinate is $z_{\rho, j}$ and whose other coordinates are fresh private variables. This added record is the *pin* at (ρ, j) .

- (iii) Make no identifications except those inherited from E and the one identification between each pin and its designated slot. In particular, distinct pin records have disjoint private variables, even when they attach to the same variable of E .

The corresponding *reversed-half lift* is obtained by reversing every coordinate of every full record in the base-half lift; the designated factor thereby moves to its mirrored copy in the reversed half. The family of pinned lifts contains both orientations and every choice of a certification coordinate in step (ii), and is therefore finite and effectively enumerable.

The pins connect the local profile types to the global type assignment.

Lemma 5.7. *Suppose an expanded signal-profile query occurs in a relation-factor projection of a target palette P using fully certified signal records. Then one of its pinned full-record lifts admits a palette homomorphism to P .*

Proof. For each selected signal record, choose an allowed full tuple of P that projects to it, and assign the coordinates outside the designated relation factor to the corresponding private variables. Since each selected signal record is fully certified, its color in every slot belongs to the corresponding set C_v^0 . For each pin, choose an allowed tuple witnessing this membership at a certification coordinate of the relevant type. The finite list contains these coordinate choices, and the remaining entries of the chosen tuples receive the private variables. $\square$

It remains to show that the lifted finite palettes are canonical palettes of ordered hypergraphs.

Canonical realizations. The basic device is the Johnson wheel. Let C_0 be an ordered set of size $K^{\text{prov}} - 2$, and let $v_0, \dots, v_{q-1}$ be distinct rim vertices. The edges $C_0 \cup \{v_{i-1}, v_i\}$, indexed by $i \in \mathbb{Z}/q\mathbb{Z}$, form a wheel.

We use the following elementary incidence property. If $q = 3$ or $q \geq 4$, the wheel edges are distinct, and two of them meet in $K^{\text{prov}} - 1$ vertices exactly when they are consecutive on the cyclic list. Their common face is $C_0 \cup \{v_i\}$ for the intervening rim vertex; every other covered face is private within the wheel. This follows because two rim pairs share a vertex exactly in the stated cases.

For the binary atoms of a fixed rail, place all rim vertices in the gap of the ordered core corresponding to the adjacent pair I_r . Order all $Y_{r,s}$ -vertices before all X_r -vertices within this gap. In an edge containing one X -rim vertex and one Y -rim vertex, deleting the earlier Y -vertex leaves the X -face and deleting the later X -vertex leaves the Y -face. Hence every A - and B -record has the common stored coordinate order (X, Y) , even though a path traverses a B -record in the reverse direction.

Map-collision queries are three-wheels. A relator path together with its closing reserve is a wheel of length at least three, because every nonempty group letter expands to two binary links. A word query whose two-link expansion has length at least four is obtained by attaching its M -record along the base face shared by the first and last records of the closed path; the written detour in (3.1) guarantees this condition for the source words.

The only coordinate calculation not covered by this common-gap construction is the cyclic M -functionality query. It is handled by the following three-wheel.

Lemma 5.8. *For each $r \in [4]$, two M -records agreeing in coordinate c_r and a reserve record joining their coordinate- $(r+1)$ colors have a canonical three-wheel realization in which the reserve occupies the pair in (5.1).*

Proof. Let C_0 be an ordered set of $K^{\text{prov}} - 2$ core vertices. Write the three edges as

$$C_0 \cup \{a, b\}, \quad C_0 \cup \{b, c\}, \quad C_0 \cup \{a, c\}.$$

The first two are the M -records; their common face is $C_0 \cup \{b\}$, and the last edge is the reserve record joining the faces $C_0 \cup \{a\}$ and $C_0 \cup \{c\}$.

For $r = 1, 2, 3$, put $j := c_r$ and $j' := c_{r+1}$. Place a, c in the gap after the first $j - 1$ core vertices and place b in the gap after the first $j' - 2$ core vertices. In each M -edge, deleting a or c has coordinate j , while deleting b has coordinate j' . In the reserve edge, a, c have ranks $j, j + 1$.

For $r = 4$, place b after the first $c_1 - 1 = 2$ core vertices and place a, c after the first $c_4 - 2 = 10$ core vertices. In each M -edge, b has rank $c_1 = 3$ and a or c has rank $c_4 = 12$. In the reserve edge, a, c have ranks 11, 12. This is precisely (5.1). The same construction works after adding the common coordinate offset of the relation block inside the full record. $\square$

We also need to attach a record at one prescribed covered face without creating other shared faces.

Lemma 5.9. *Let $(H, <)$ be an ordered K^{prov} -graph, let f be a covered $(K^{\text{prov}} - 1)$ -face of H , and let $j \in [K^{\text{prov}}]$. There is an extension of the vertex order and a fresh vertex y such that, after adding the edge $f \cup \{y\}$,*

- (a) *the face f occurs in coordinate j of the new canonical record;*
- (b) *every other face of the new edge is new and private; and*
- (c) *the new edge meets an old edge in $K^{\text{prov}} - 1$ vertices exactly when that old edge already contains f .*

More generally, fix a finite list $f_1, \dots, f_t$ of covered faces of the original graph H , with repetitions allowed. One new edge can be attached along each f_i , using pairwise distinct new vertices, without creating any shared face other than the prescribed attachment faces.

Proof. Place y in the j th gap of the ordered set f , so exactly $j - 1$ vertices of f precede y . Then deleting the vertex of rank j , namely y , leaves f . Every other face of $f \cup \{y\}$ contains the fresh vertex y , so it is not a face of an old edge. Moreover, for an old edge e , the intersection $e \cap (f \cup \{y\})$ equals $e \cap f$; it has size $K^{\text{prov}} - 1$ exactly when $f \subseteq e$. For the fixed list of original faces, insert the fresh vertices one at a time. The required gap is determined only by the order on the vertices of the corresponding original face, and distinct fresh vertices may be ordered arbitrarily inside the same ambient interval. Two attached edges on distinct original faces meet in at most $K^{\text{prov}} - 2$ old vertices, while two attached edges on the same original face share exactly that face. $\square$

The exact incidence schemes for the four query classes are summarized in Table 1. Here a *link record* means one binary $A_{r,s}$ - or $B_{r,s}$ -record after profile expansion, and z denotes the written test word currently being compiled; $|z|$ is its number of letters before the two-link expansion.

Query class	Core records	Core incidence scheme	Prescribed coordinates
Map collision	two map records and one reserve	a three-wheel	the map-map face is the repeated U - or V -variable; the other two faces occupy the corresponding slots of I_r in the map and reserve profiles
Relator q	$2 q $ link records and one closing reserve	a $(2 q + 1)$ -wheel, hence a wheel of length at least three	consecutive link faces alternate between the X_r - and $Y_{r,s}$ -slots in I_r ; the reserve closes the two X_r endpoints
Cyclic M -functionality	two M -records and one reserve	a three-wheel	the M -records share coordinate c_r ; their coordinate- c_{r+1} faces meet the reserve in I_{r+1} , using (5.1) when $r = 4$
Word test for z	$2 z $ link records and one M -record	a $2 z $ -wheel together with one additional record attached along the base face shared by the first and last wheel records	the wheel uses I_r ; its base X_r -face is attached to coordinate c_r of the additional M -record

Table 1: Core incidences in the canonical query construction.

The table contains every equality prescribed by Definitions 4.2 and 5.3 and no others. Indeed, for binary records, place the rim vertices in the gap assigned to I_r and put each $Y_{r,s}$ -vertex before each X_r -vertex. A map collision closes after three records, and a relator q gives a wheel of length $2|q| + 1 \geq 3$. The cyclic- M row is Lemma 5.8. For the word row, assume that the two-link expansion of z has length at least four, identify its endpoints, and attach the M -record at the base face using Lemma 5.9. In particular, $z = w^\dagger(e)$ satisfies this assumption by (3.1).

We can now realize every pinned lift by one ordered hypergraph.

Proposition 5.10. *For every choice of provisional parameters R , $(m_v)_{v \in \mathcal{T}}$, and h , every expanded query, every relation-factor copy, every half of the resulting record, and every choice of certification coordinates*

in [Definition 5.6](#), one can effectively construct an ordered K^{prov} -graph $(J, <)$ whose canonical palette is isomorphic to the corresponding pinned full-record lift. For a word query, assume that its two-link expansion has length at least four. Under this isomorphism, every prescribed equality of record variables is represented by one common covered $(K^{\text{prov}} - 1)$ -face, and every unprescribed slot is represented by a private face. In particular, no two distinct records share more than one palette color.

Proof. We construct first the records belonging to the unpinned query and then attach all pins.

The core incidence scheme. Use the row of [Table 1](#) corresponding to the query. Every core component is a wheel of length three or at least four, possibly with the additional face-attached M -record in the word-test row, and its shared faces are exactly the variables identified by the expanded query.

For a binary wheel, use one common ordered $(K^{\text{prov}} - 2)$ -set and place the rim vertices in the gap assigned to the relevant adjacent coordinate pair, with all auxiliary-type rim vertices before all master-type rim vertices. The rank calculation preceding [Lemma 5.8](#) puts the two stored colors in exactly the prescribed coordinates. The two rim vertices of the unique reserve edge have the same type; choose their relative order inside that type block to realize the selected reserve orientation. For a cyclic- M core, use [Lemma 5.8](#); when the oriented reserve profile is fixed, choose the relative order of the two same-gap rim vertices to realize that orientation. For the face-attached M -record in a word query, apply [Lemma 5.9](#) at the full-record coordinate corresponding to the local position c_r , so the shared base face occupies the required M -coordinate even though it generally occupies a different coordinate in the adjacent binary record. Coordinate offsets coming from the location of the relation factor inside the full record are added throughout. This constructs the core for the base-half lift; the reversed-half lift is handled by a global order reversal after all pins have been attached.

By the wheel incidence observation above, the wheel edges have precisely the common faces prescribed by the cyclic query variables and no others. [Lemma 5.9](#) gives the same conclusion for the optional face-attached M -record. Hence, at this stage, the covered faces are in bijection with the variables of the unpinned full-record pattern. The shared variables are the specified wheel faces and all remaining record slots are private.

Pins and global order. For every one of the L slots in the designated signal-profile factor of each core record, let f be the face that represents its color variable. Attach a fresh edge along f at the chosen certification coordinate, using [Lemma 5.9](#). If the same query variable occurs in several core records, all corresponding pins attach to the same face; this creates exactly the identifications prescribed by the pinned lift. All nonshared slots of every pin record are private. Successive face attachments extend, rather than alter, the order already placed on the core, so they produce one global linear order on $V(J)$.

For a lift in the reversed half, first identify that half with a base half by reversing its coordinate labels and carry out the preceding construction, including all pins. Then equip the same hypergraph with the reverse global vertex order $<_{\text{rev}}$. Indeed, if $e = \{v_1 < \dots < v_{K^{\text{prov}}}\}$, then its j th canonical face under $<_{\text{rev}}$ is $e \setminus \{v_{K^{\text{prov}}+1-j}\}$. Consequently $\text{Can}(J, <_{\text{rev}}) = \text{rev Can}(J, <)$. Reversal simultaneously mirrors the factor offsets, the orientation of every reserve profile, the cyclic M -coordinates, the word-wheel and face-attached M -record coordinates, and all pin coordinates. The underlying edge intersections are unchanged.

It remains only to exclude unintended identifications. Two core edges have the wheel intersections described above. An attached edge can share a $(K^{\text{prov}} - 1)$ -face with another edge only through its attachment face, by [Lemma 5.9](#). Therefore every common covered face is one of the query equalities or pin equalities already listed, and every other face occurs once. This proves exactness and also shows that two distinct records share at most one color.

All vertex sets, gap choices, edge lists, and order extensions are obtained by finite explicit operations from the expanded query. The construction is therefore effective. $\square$

6 Density bound and rational calibration

The constructions in [Lemma 5.5](#) and [Proposition 5.10](#), followed by the outer-order construction, produce ordinary finite forbidden K^{prov} -graphs. We now bound the density of any palette deficient for these graphs and choose the parameters so that the bound has a prescribed rational maximizer.

6.1 The negative relation-factor bound

Assume throughout that the tested word has property (W). We convert the finite-action dichotomy into a bound for any target palette satisfying the compiled local constraints; certification removes all uncertified colors from the final expression.

Fix one half-record and one relation-factor copy. Let Q be a repetition-free target palette of the provisional full-record length K^{prov} , equipped with a probability distribution μ and a homomorphism $Q \rightarrow B_A$. For each signal type word $\sigma = (\sigma_1, \dots, \sigma_L)$, let R_σ be the restriction of the factor projection to the corresponding signal box and put

$$R_\sigma^0 := R_\sigma \cap \prod_{j=1}^L C_{\sigma_j}^0.$$

Thus R_σ^0 consists exactly of the fully certified signal records in that box. Assume that the collection $(R_\sigma^0)_{\sigma \in \Sigma}$ contains no expanded query. Let $\mathbf{a} = (a_v)_{v \in \mathcal{T}}$ be the vector of total type masses, and let $\mathbf{u}$ be the certified type-mass vector from (5.4).

Extract $\widehat{D}$, N_{D,U_D} , N_{D,V_D} , and M from the active supports of the relations R_σ^0 , as in Lemma 5.4. Put

$$e_D^0 := h(\widehat{D}) + n(N_{D,U_D}) + n(N_{D,V_D}), \quad \delta_D := b_D(\mathbf{u}) - e_D^0.$$

Write $\boldsymbol{\delta} = (\delta_D)_{D \in \mathcal{D}}$. By Lemma 5.4, the extracted typed structure avoids all abstract queries. Thus (4.2) gives $\delta_D \geq 0$, and property (W) together with Theorem 4.5 gives

$$m(M) \leq \kappa \sum_{D \in \mathcal{D}} \delta_D. \quad (6.1)$$

The cancellation of the unrestricted part of the factor is the reason for certifying every coordinate of a signal profile. For a signal type word σ , write a^σ and u^σ for the corresponding coordinate monomials. If we enlarge the projection by allowing every off-signal tuple and every signal tuple touching an uncertified color, their combined density is at most

$$\left(1 - \sum_{\sigma \in \Sigma} a^\sigma\right) + \sum_{\sigma \in \Sigma} (a^\sigma - u^\sigma) = 1 - \sum_{\sigma \in \Sigma} u^\sigma. \quad (6.2)$$

Thus the total type masses a_v disappear from the upper bound.

Let $\chi(\mathbf{u}) := \prod_{r=1}^4 u_{X_r}$. Apply (5.2) to the fully certified relations R_σ^0 . Their tag-and-filler coordinate classes have masses u_v . For a package D , the full capacity of its four certified signal boxes and the bound on their restricted contribution from (5.3) are, respectively,

$$\begin{aligned} &\theta_D(\mathbf{u})(2u_{U_D}u_{V_D} + u_{U_D}^2 + u_{V_D}^2), \\ &2\theta_D(\mathbf{u})e_D^0 = \theta_D(\mathbf{u})(u_{U_D}^2 + u_{V_D}^2 - 2\delta_D). \end{aligned}$$

The M -box has capacity $\beta(\mathbf{u})\chi(\mathbf{u})$ and restricted contribution at most $\beta(\mathbf{u})m(M)$. Substituting (6.1) gives the following factor bound.

Proposition 6.1. *Let Q be a repetition-free target palette of record length K^{prov} , equipped with a probability distribution and a homomorphism $Q \rightarrow B_A$. Fix a half-record and one relation-factor projection, and define $\mathbf{u}$, $\widehat{D}$, N_{D,U_D} , N_{D,V_D} , M , and $\boldsymbol{\delta}$ as above. If the tested word has property (W) and no fully certified expanded query occurs in this projection, then its weighted density is at most*

$$S_-(\mathbf{u}, \boldsymbol{\delta}) := 1 - \sum_{D \in \mathcal{D}} 2\theta_D(\mathbf{u})u_{U_D}u_{V_D} - \beta(\mathbf{u})\chi(\mathbf{u}) - \sum_{D \in \mathcal{D}} (2\theta_D(\mathbf{u}) - \kappa\beta(\mathbf{u}))\delta_D.$$

Moreover, $S_-(\mathbf{u}, \boldsymbol{\delta}) \geq 0$ for every parameter tuple extracted from such a projection.

Proof. Combining (6.2) with the restricted signal contributions gives the intermediate upper bound

$$E := 1 - \sum_D 2\theta_D(\mathbf{u})u_{U_D}u_{V_D} - \beta(\mathbf{u})\chi(\mathbf{u}) - \sum_D 2\theta_D(\mathbf{u})\delta_D + \beta(\mathbf{u})m(M).$$

The decomposition into off-signal or uncertified capacity and restricted signal contributions shows directly that $E \geq 0$. Applying (6.1) gives $E \leq S_-(\mathbf{u}, \boldsymbol{\delta})$, and therefore the latter is also nonnegative for every parameter tuple extracted from a target. $\square$

A rational interior point. We now choose a rational probability vector $\mathbf{p} = (p_v)_{v \in \mathcal{T}}$ with the following properties.

- (i) all coordinates of $\mathbf{p}$ are positive;
- (ii) all action types have the same mass; and
- (iii)

$$\kappa\beta(\mathbf{p}) < 2 \min_{D \in \mathcal{D}} \theta_D(\mathbf{p}). \quad (6.3)$$

The choice can be made effectively. Before normalization, assign weight 1 to every action type and to $\mathbf{F}, \mathbf{F}_0, \mathbf{F}_1$, weight t to every $\mathbf{Tag}_D$, and weight t^3 to $\mathbf{Tag}_M$. For rational $t > 0$, normalize by the rational total mass. As $t \downarrow 0$, the ratio $\beta(\mathbf{p})/\theta_D(\mathbf{p})$ is $O(t^2)$ uniformly in D , proving (6.3) for small enough t . At the same time, the total capacity of all signal boxes tends to zero. We may therefore require in addition that

$$\tau_0 := S_-(\mathbf{p}, \mathbf{0}) = 1 - \sum_D 2\theta_D(\mathbf{p})p_{U_D}p_{V_D} - \beta(\mathbf{p})\chi(\mathbf{p}), \quad 0 < \tau_0 < 1. \quad (6.4)$$

All these conditions are strict rational inequalities and can be checked by a finite search over positive rational t .

6.2 Effective rational calibration

We now choose the certification multiplicities so that the rational point $\mathbf{p}$ is the unique maximizer of the preceding upper bound. This step is an entropy maximization on the simplex.

A projection inequality. Let $R \subseteq C^I$ be a relation on a finite set C with probability measure μ . If $I = J_1 \sqcup \dots \sqcup J_t$ is a partition, then $R \subseteq \prod_{j=1}^t \text{proj}_{J_j} R$. Taking product measure gives

$$d_\mu(R) \leq \prod_{j=1}^t d_\mu(\text{proj}_{J_j} R). \quad (6.5)$$

No rectangularity assumption on R is involved.

For provisional parameters $R \geq 1$ and $m_v \geq 1$, use R copies of the relation factor, each with its own expanded query list. In the negative case, whenever no expanded query occurs on fully certified signal records, the projection of an arbitrary target onto one half of the record is bounded, by (5.5), Proposition 6.1, and (6.5), by

$$\mathcal{U}_{R,m}(\mathbf{u}, \boldsymbol{\delta}^{(1)}, \dots, \boldsymbol{\delta}^{(R)}) := \left(\prod_{v \in \mathcal{T}} u_v^{m_v} \right) \prod_{\ell=1}^R S_-(\mathbf{u}, \boldsymbol{\delta}^{(\ell)}).$$

The relevant compact *semialgebraic domain*, meaning a set defined by finitely many polynomial equalities and inequalities, is

$$\begin{aligned} \mathcal{D} := \{(\mathbf{u}, \boldsymbol{\delta}^{(1)}, \dots, \boldsymbol{\delta}^{(R)}) : & u_v \geq 0, \quad \sum_v u_v \leq 1, \\ & 0 \leq \delta_D^{(\ell)} \leq b_D(\mathbf{u}) \quad (D \in \mathcal{D}, \ell \in [R]), \\ & S_-(\mathbf{u}, \boldsymbol{\delta}^{(\ell)}) \geq 0 \quad (\ell \in [R])\}. \end{aligned} \quad (6.6)$$

Every parameter tuple extracted from a target lies in $\mathcal{D}$.

We now specialize these provisional layout parameters. Put $S(\mathbf{u}) := S_-(\mathbf{u}, \mathbf{0})$ and define $g_v := \frac{\partial}{\partial u_v} \log S(\mathbf{u}) \Big|_{\mathbf{u}=\mathbf{p}}$. These numbers are rational because S has rational coefficients and $S(\mathbf{p}) = \tau_0 > 0$. Choose a positive integer R such that $p_v R g_v \in \mathbb{Z}$ for every v . For each positive multiple N_{cal} of a common denominator of the p_v , set

$$m_v := p_v(N_{\text{cal}} - R g_v).$$

Call N_{cal} *admissible* when every m_v is positive. For every sufficiently large such multiple, N_{cal} is admissible and all m_v are positive integers.

The following proposition makes this choice effective and identifies the unique maximizing type distribution.

Proposition 6.2. *There is an effectively computable admissible value of N_{cal} for which*

$$\mathcal{U}_{R,m}(\mathbf{u}, \boldsymbol{\delta}^{(1)}, \dots, \boldsymbol{\delta}^{(R)}) \leq T_0 \quad \text{on } \mathcal{D}, \quad (6.7)$$

where

$$T_0 := \left(\prod_{v \in \mathcal{T}} p_v^{m_v} \right) \tau_0^R \in \mathbb{Q}.$$

Moreover, equality in (6.7) is possible only when $\mathbf{u} = \mathbf{p}$ and $\boldsymbol{\delta}^{(1)} = \dots = \boldsymbol{\delta}^{(R)} = \mathbf{0}$.

Proof. We first prove existence for all sufficiently large admissible N_{cal} . Fix one admissible value N_0 , and write $\mathcal{U}_N, T_0(N)$ for the quantities corresponding to $N = N_{\text{cal}}$. On the interior of the positive orthant,

$$\frac{\mathcal{U}_N}{T_0(N)} = \frac{\mathcal{U}_{N_0}}{T_0(N_0)} W(\mathbf{u})^{N-N_0}, \quad W(\mathbf{u}) := \prod_{v \in \mathcal{T}} \left(\frac{u_v}{p_v} \right)^{p_v}. \quad (6.8)$$

This identity is first read on the positive orthant. For $N > N_0$, both sides extend continuously to the boundary, with value zero whenever some $u_v = 0$, because all certification exponents are positive. Weighted AM–GM yields

$$W(\mathbf{u}) \leq \sum_v p_v \frac{u_v}{p_v} = \sum_v u_v \leq 1.$$

Equality throughout holds only at $\mathbf{u} = \mathbf{p}$. Hence outside any fixed neighborhood of $\mathbf{p}$, compactness gives $W(\mathbf{u}) \leq 1 - \eta$ for some $\eta > 0$. The normalized function for N_0 is bounded on $\mathcal{D}$, so (6.8) proves the strict inequality in (6.7) outside that neighborhood for all sufficiently large N .

Near $\mathbf{p}$, condition (6.3) remains strict. Shrink the neighborhood so that every u_v and $S(\mathbf{u})$ are positive there. Since S_- is affine in every deficit variable, $S_-(\mathbf{u}, \boldsymbol{\delta}) \leq S(\mathbf{u})$ throughout this neighborhood, with equality only when $\boldsymbol{\delta} = \mathbf{0}$. It remains to maximize the zero-deficit expression. Its logarithm is

$$\sum_v m_v \log u_v + R \log S(\mathbf{u}) = N \sum_v p_v \log u_v + R \left(\log S(\mathbf{u}) - \sum_v p_v g_v \log u_v \right). \quad (6.9)$$

The second term has zero gradient at $\mathbf{u} = \mathbf{p}$ by the definition of g_v . Moreover, in a fixed neighborhood of $\mathbf{p}$,

$$\sum_v p_v \log u_v \leq \sum_v p_v \log p_v + \sum_v (u_v - p_v) - c \|\mathbf{u} - \mathbf{p}\|^2$$

for some $c > 0$. The second term of (6.9) differs from its value at $\mathbf{p}$ by at most $C \|\mathbf{u} - \mathbf{p}\|^2$. Since $\sum_v (u_v - p_v) \leq 0$ on the simplex, choosing $N > C/c$ proves that $\mathbf{p}$ is the unique constrained maximizer in this neighborhood. This completes the existence proof.

For effectivity, enumerate the admissible multiples N_{cal} . For each one, use real quantifier elimination to check both (6.7) and the assertion that equality forces $\mathbf{u} = \mathbf{p}$ and all deficits to vanish. These are first-order sentences over the real closed field with rational coefficients. The domain (6.6) is compact semialgebraic and the objective is a polynomial because all exponents are nonnegative integers. Quantifier elimination is an algorithm that decides such first-order statements over the real numbers; see [2, Chapter 14]. The existence argument guarantees that the search terminates. $\square$

The palindromic record. The base half-record has $K_0 := \sum_{v \in \mathcal{T}} m_v + RL$ coordinates. Its intended relation is the Cartesian product of the prescribed certification-coordinate classes and the R relation-factor relations. The type shell is rectangular. A certification coordinate for v allows every color of type v , while every relation coordinate allows every type.

Set

$$K_* := 2K_0 \geq 4, \quad T := T_0^2 \in \mathbb{Q}.$$

These two constants are fixed and effectively computable, independently of the input word. Moreover, $0 < T < 1$. Positivity follows from (6.4), while $T_0 < 1$ because every m_v is positive and $\mathbf{p}$ has at least two positive coordinates.

We now fix any $K \geq K_*$ and put $h := K - K_*$. In a realization with color set C , take the Cartesian product of the base relation, the full relation C^h , and the coordinate reverse of the base relation. We call the result the *palindromic record*; an allowed full record has the form

$$(a_1, \dots, a_{K_0}, z_1, \dots, z_h, b_{K_0}, \dots, b_1), \quad a, b \text{ allowed base records, } z_1, \dots, z_h \in C. \quad (6.10)$$

Thus the h middle coordinates are completely unrestricted and contribute a factor 1 to every weighted density. Their shell sets are all $\mathcal{T}$. The full intended palette is invariant under coordinate reversal. Reversal exchanges the two base relations and reverses the unrestricted middle block. Its rectangular type shell is palindromic as well, since

$$A_i = A_{K+1-i} \quad (i \in [K]), \quad \text{and hence} \quad \text{rev } B_A = B_A. \quad (6.11)$$

Consequently, if a canonical shell star does not map to B_A , neither does its reversed canonical palette. Indeed, a map $\text{rev Can}(J) \rightarrow B_A$ would reverse to a map $\text{Can}(J) \rightarrow \text{rev } B_A = B_A$.

For an arbitrary target palette, use the shell obstruction to obtain one global type assignment. In the reversed half we first reverse the coordinate order, so its profiles and extracted relations are identified with those of a base half. The certified color sets in the two halves may nevertheless be different; denote the resulting parameters by $(\mathbf{u}^+, \delta^+)$ and $(\mathbf{u}^-, \delta^-)$. Partition the coordinates into the two base halves and the middle block. The middle projection has weighted density at most 1, so (6.5) and Proposition 6.2, applied separately to the two halves, give

$$p_Q(\mu) \leq \mathcal{U}_{R,m}(\mathbf{u}^+, \delta^+) \cdot 1 \cdot \mathcal{U}_{R,m}(\mathbf{u}^-, \delta^-) \leq T_0^2 = T \quad (6.12)$$

whenever the negative-case constraints hold in both halves.

7 Finite forbidden families and undecidability

The constants K_* and T are now fixed. We construct the finite families for an arbitrary chosen $K \geq K_*$, prove the density dichotomy uniformly in K , and then deduce the two undecidability theorems from the Introduction.

The finite forbidden families. Fix $K \geq K_*$. The type set, the two base halves, the shell, the number of factor copies, the certification multiplicities, and the $K - K_*$ unrestricted middle coordinates are now fixed. Only the written word query depends on the source input.

For any finite written word z over $S_0^{\pm 1}$ whose two-link expansion has length at least four, define a finite family $\mathcal{F}_K[z]$ of ordinary K -graphs as follows. In the word query and its canonical wheel construction, replace $w^\dagger$ throughout by this chosen word z ; the length assumption is exactly the one used in the word row of Table 1.

- (1) Include $\mathfrak{D}(J_I)$ for every ordered shell-star realization $J_I \in \mathcal{S}_A$ from Lemma 5.5.
- (2) For every relation-factor copy and each half of the palindromic record, include $\mathfrak{D}(J)$ for every canonical realization $(J, <)$ of a pinned full-record lift of every expanded map-collision, relator, cyclic- M , and z -word query.

In the second item, all map-copy choices, reserve orientations, reserve-package tags, and certification-coordinate choices are expanded. Every unrestricted middle slot is private. The family is finite. Its cardinality and the orders of its members may grow with K and $|z|$, but its uniformity is K . For $e \in \mathbb{N}$, put $\mathcal{F}_{K,e} := \mathcal{F}_K[w^\dagger(e)]$. This is defined because the written detour in (3.1) makes the two-link expansion have length at least four.

The function $(K, e) \mapsto \mathcal{F}_{K,e}$, with domain $K \geq K_*$, is computable in the encoding fixed in Section 2. To compute it, first obtain $w^\dagger(e)$ from the fixed group-theoretic source. The presentation, type set, factor copies, and certification coordinates are fixed finite data; the unrestricted middle block is determined by $K - K_*$. Enumerate all profile expansions, pin choices, and shell stars, apply Proposition 5.10, and find each required ordered design by the terminating search in Lemma 5.1. Every remaining operation is a finite enumeration.

The construction has the following density dichotomy.

Theorem 7.1. *For every $K \geq K_*$ and every $e \in \mathbb{N}$,*

$$\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) \geq T. \quad (7.1)$$

Moreover,

$$\begin{aligned} e \in \mathcal{N} &\implies \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) = T, \\ e \in \mathcal{Y} &\implies \pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) > T. \end{aligned}$$

Proof. Fix $K \geq K_*$. We divide the proof into three parts.

Negative upper bound. Assume $e \in \mathcal{N}$. By [Theorem 3.1](#), the tested word $w^\dagger(e)$ has property (W). We first prove that every repetition-free $\mathcal{F}_{K,e}$ -deficient palette Q , under every probability distribution on its colors, has value at most T .

Deficiency with respect to the shell obstructions and [Lemma 5.5](#) gives a global type assignment $Q \rightarrow B_A$. Fix a relation-factor copy in one half. If an expanded typed query occurred on fully certified signal records, [Lemma 5.7](#) would provide a pinned full-record lift mapping to Q . By [Proposition 5.10](#) and [Lemma 5.2](#), its ordinary outer-order graph would be Q -colorable, contradicting deficiency. Thus the extracted union supports satisfy all negative-case constraints. Applying [Propositions 6.1](#) and [6.2](#) in every factor and then [\(6.12\)](#) gives $p_Q(\mu) \leq T$.

Now let P be an arbitrary deficient palette and let $\mathbf{x}$ be any probability vector on its colors. By [Lemma 2.3](#), there are repetition-free deficient palettes P_N whose unweighted densities tend to $p_P(\mathbf{x})$. Applying the preceding bound with the uniform distribution on P_N gives $p_P(\mathbf{x}) \leq T$. Hence $\Lambda(P) \leq T$ for every deficient palette, and [Proposition 2.2](#) yields $\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) \leq T$.

Lower bound. Construct the intended palindromic palette using the rational type masses $\mathbf{p}$. For every type $v \in \mathcal{T}$ —including all action, tag, and filler types—use one color of weight p_v . This lower-bound witness need not be repetition-free. Use the unique map edge in both copies of every link, empty loopless reserves, and $M = \emptyset$. Allow every color of the prescribed type in each certification coordinate, and complete all off-signal boxes and the unrestricted middle block. The base-half relation is the Cartesian product of its certification-coordinate classes and relation-factor relations, and the full relation is the product of the base relation, the unrestricted middle block, and the coordinate-reversed base relation, as in [\(6.10\)](#). By [Lemma 4.6](#), every package attains its target value and every local query is absent. Each relation factor has value exactly τ_0 , each base half has value T_0 , and the middle block has value 1. Hence the full palette has value $T = T_0^2$.

This palette is $\mathcal{F}_{K,e}$ -deficient. Indeed, it maps to the type shell, so no minimal empty-intersection canonical star maps to it. By [\(6.11\)](#), the reversed canonical star cannot map either. In any homomorphism from a pinned lift, each pin forces its shared color to have the prescribed global type; hence every core record must lie in its intended signal box. The middle slots are private and impose no further condition. Therefore no canonical query lift maps to the palette because the underlying signal query is absent. If a reversed canonical lift mapped to it, then the unreversed lift would map to the reverse palette; the latter equals the original palindromic palette. Thus both branches of [Lemma 5.2](#) are excluded for shell obstructions and query lifts.

The weighted palette transfer gives [\(7.1\)](#). Together with the negative upper bound, this proves the first implication in [Theorem 7.1](#).

Strict lower bound. Assume $e \in \mathcal{Y}$ and choose a finite quotient H in which $w^\dagger(e) \neq 1$. On every rail use the right regular action from [Lemma 4.7](#), distributing each action-type mass uniformly over its copy of H . Complete all reserves, certification-coordinate classes, off-signal boxes, and unrestricted middle coordinates, and use the diagonal four-matching in every relation factor. The palette maps to B_A , so no minimal empty-intersection canonical star maps to it; by [\(6.11\)](#), no reversed canonical star maps to it either. If a pinned lift mapped to this palette, its pins would force every core color into the prescribed type class. After forgetting the profile choices and the private tag, filler, off-factor, and middle coordinates, the core records would therefore give an occurrence of the corresponding abstract typed query. Hence no canonical query lift maps to the palette. If a reversed canonical lift mapped to it, then the unreversed lift would map to the reverse palette; the latter equals the original palette by the palindromic construction. Thus both branches of [Lemma 5.2](#) are excluded, and the palette is $\mathcal{F}_{K,e}$ -deficient.

Every package remains at its target value, while each relation factor gains the strictly positive term $\beta(\mathbf{p})m(M)$, so its value is greater than τ_0 . Consequently each half has value greater than T_0 , the middle block has value 1, and the full palette has value greater than T . The weighted transfer proves the second implication in [Theorem 7.1](#). $\square$

The decision reduction. We first prove the rational-threshold statement.

Proof of Theorem 1.1. Fix $K \geq K_*$ and any one of the relations $=$, $\leq$, and $>$. Suppose that an algorithm decided that relation between $\pi_{K-2}^{\text{unif},(K)}(\mathcal{F})$ and T on inputs satisfying the stated promise. On input e , compute $\mathcal{F}_{K,e}$ and run the algorithm; the promise and the two outcomes follow from [Theorem 7.1](#). On inputs $e \in \mathcal{N}$, the predicates $=$ and $\leq$ hold and $>$ fails; on inputs $e \in \mathcal{Y}$, the reverse holds. No assertion is needed outside $\mathcal{Y} \cup \mathcal{N}$. For $>$, the accepting set of the algorithm is therefore a recursive set containing $\mathcal{Y}$ and disjoint from $\mathcal{N}$; for $=$ or $\leq$, its complement has that property. Either case recursively separates $\mathcal{Y}$ from $\mathcal{N}$, a contradiction. $\square$

To turn the rational threshold into a fixed finite comparison family, fix $K \geq K_*$, use the same construction with the word $w_0 := s_0 s_0^{-1}$, and put $\mathcal{G}_K := \mathcal{F}_K[w_0]$. The word w_0 has property (W). The statements and proofs of [Lemma 5.4](#) and [Proposition 6.1](#) and the negative upper-bound argument use the source input only through this property. They therefore apply verbatim to w_0 ; together with the universal equality witness, they give

$$\pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K) = T. \quad (7.2)$$

Proof of Theorem 1.2. Fix $K \geq K_*$. For every input e , (7.1) and (7.2) give the promised inequality

$$\pi_{K-2}^{\text{unif},(K)}(\mathcal{F}_{K,e}) \geq \pi_{K-2}^{\text{unif},(K)}(\mathcal{G}_K).$$

On inputs $e \in \mathcal{N}$, equality holds, while on inputs $e \in \mathcal{Y}$ the inequality is strict; no assertion is needed outside $\mathcal{Y} \cup \mathcal{N}$. As in the proof of [Theorem 1.1](#), a decision algorithm for any one of the three relations, complemented when necessary, would yield a recursive set containing $\mathcal{Y}$ and disjoint from $\mathcal{N}$, contrary to recursive inseparability. $\square$

Thus one rational threshold works for every $K \geq K_*$; we make no claim for smaller K , including $K = 3$.

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Declaration on the use of AI

The authors used generative AI tools to discuss the organization of the proofs, check proof details, and improve the exposition. The authors take full responsibility for all mathematical claims and proofs.

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